

Composite v0.4 — Term 2 Substrate Decomposition $N_c^4 / 2^{N_c}$

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2026-06-04 Thursday ~14:25 EDT

Composite v0.4 — Term 2 Substrate Decomposition

0. Goal

Per Composite v0.3 (~13:42 EDT) + Self-Audit Cal #35 (~13:50 EDT): Term 2 $N_c^4 / 2^{N_c} = 81/8 = 10.125$ substrate-natural identified as form-fitting candidate per Cal #35 STANDING. This v0.4 examines whether Term 2 substrate-natural form decomposes via substrate-Mechanism FORCING candidates per Casey #14 STANDING.

1. Term 2 Numerical Decomposition

$$\frac{N_c^4}{2^{N_c}} = \frac{3^4}{2^3} = \frac{81}{8} = 10.125$$

Substrate-natural factor identification candidates: - Numerator: N_c^4 substrate-natural = (substrate-color)^(substrate-natural exponent 4) - Denominator: 2^{N_c} substrate-natural = (substrate-rank or substrate-Cartan-base)^(substrate-color) - Substrate-natural exponent 4 candidate: $\text{codim}(\text{SO}(3,1) \subset \text{SO}(4,2) \subset \text{SO}(5,2))$ per Casey #14 STANDING substrate-mechanism class

2. Substrate-Mechanism FORCING Candidate per Casey #14 STANDING

Per Casey #14 STANDING (Substrate-Predicted 3+1 Minkowski Signature via $1/n_c$ chirality projection):

Substrate-codim-4 substrate-mechanism: $4 = \text{codim}(\text{SO}(3,1) \subset \text{SO}(5,2))$ per substrate-Casey-#14-codim-4 substrate-mechanism class.

Substrate-codim-4 substrate-natural exponent candidate: substrate-codim-4 substrate-mechanism produces substrate-natural exponent 4 per Casey #14 STANDING substrate-mechanism cascade.

3. Term 2 Substrate-Decomposition Candidate v0.4

$$\frac{N_c^4}{2^{N_c}} = \frac{N_c^{\text{codim}(\text{SO}(3,1) \subset \text{SO}(5,2))}}{2^{N_c}}$$

Substrate-mechanism FORCING reading candidate: substrate-color N_c raised to substrate-Casey-#14-codim-4 substrate-natural power, divided by substrate-Cartan-base 2 raised to substrate-color N_c substrate-natural power.

Substrate-natural factor	Substrate-mechanism FORCING candidate
$3 = N_c$	substrate-color FORCING
$4 = \text{codim}(\text{SO}(3,1) \subset \text{SO}(5,2))$	substrate-Casey-#14-codim-4 FORCING

Substrate-natural factor	Substrate-mechanism FORCING candidate
2 = substrate-Cartan-base (rank or substrate-natural 2)	substrate-rank or substrate-Cartan FORCING
3 = N_c (exponent in 2^{N_c})	substrate-color FORCING

Substantive observation v0.4: each substrate-natural factor in Term 2 $N_c^4 / 2^{N_c}$ has a substrate-mechanism FORCING candidate per Casey-named principles substrate-mechanism class.

4. Cal #35 STANDING Honest Verdict v0.4

Per Cal #35 STANDING (Independence-Taxonomy-Before-Multiplicative-Null-Model + form-selection $\neq$ mechanism-forcing):

Honest verdict v0.4: - substrate-natural factor identification per Casey #14 STANDING substrate-codim-4 substrate-mechanism is substantive substrate-mechanism candidate IDENTIFICATION - substrate-mechanism FORCING form DERIVATION (i.e., substantive proof that substrate-mechanism class FORCES this specific Term 2 form at gen-2 $V_{(3/2, 1/2)}$) is multi-week per Cal #189 - substrate-natural integer composition is SUGGESTIVE substantive substrate-mechanism candidate IDENTIFICATION; NOT substrate-mechanism FORCING form derivation

Substantive promotion candidate v0.4: Term 2 substrate-natural form may promote from form-fitting (v0.3) to PARTIAL substrate-mechanism FORCING candidate (v0.4) pending multi-week explicit substrate-mechanism derivation per Casey #14 STANDING substrate-codim-4 + substrate-color N_c substrate-mechanism class per Cal #189.

5. Composite v0.4 Substantive State

Term	Form	Status v0.4
Term 1	$n_C \cdot (n_C/\text{rank})_3 = 196.875$	substantive substrate-mechanism FORCING candidate via substrate-Bergman + substrate-Pochhammer Schur II multi-week
Term 2	$N_c^4 / 2^{N_c} = 10.125$	PARTIAL substrate-mechanism FORCING candidate via Casey #14 STANDING substrate-codim-4 + substrate-color N_c substrate-mechanism class multi-week
Composite	$n_C \cdot (5/2)_3 + N_c^{4/2} N_c = 207$	substantive 0.112% Tier 1/Tier 2 boundary precision vs observed $m_\mu/m_e \approx 206.768$

6. Substantive Multi-Week Per Cal #189

Term 2 substrate-mechanism FORCING form derivation multi-week candidate steps: - Step T2-1: substrate-codim-4 substrate-mechanism cascade derivation per Casey #14 STANDING substrate-mechanism class - Step T2-2: substrate-color N_c substrate-natural exponent derivation at substrate-Casey-#14-codim-4 substrate-mechanism class - Step T2-3: substrate-Cartan-base 2 substrate-natural denominator derivation per substrate-Cartan substrate-mechanism - Step T2-4: substrate-natural exponent N_c in denominator 2^{N_c} substrate-color substrate-mechanism

7. Closure v0.4

Composite v0.4 Term 2 Substrate-Decomposition:

Substantive observation: Term 2 $N_c^4 / 2^4 N_c = 81/8$ substrate-natural form decomposes via substrate-Casey-#14-codim-4 substrate-natural exponent + substrate-color N_c substrate-natural exponent + substrate-Cartan-base 2 substrate-natural form.

Substantive promotion: Term 2 may promote from form-fitting (v0.3) to PARTIAL substrate-mechanism FORCING candidate (v0.4) pending multi-week explicit derivation per Cal #189.

Honest framing per Cal #35 STANDING: substrate-natural factor IDENTIFICATION $\neq$ substrate-mechanism FORCING form DERIVATION; multi-week explicit FORCING derivation per Cal #189 distinguishes.

Per Cal #35 STANDING + Cal #36 STANDING + Cal #27 STANDING + Cal #189: Term 2 substrate-mechanism FORCING candidate v0.4 identified via substrate-Casey-#14-codim-4 substrate-mechanism class cross-link; substantive multi-week explicit derivation per Cal #189 required for FORCING-tier promotion.

Tier: Term 2 PARTIAL substrate-mechanism FORCING candidate v0.4; substantive multi-week explicit FORCING derivation per Cal #189.

— Lyra, Thu 2026-06-04 ~14:25 EDT. Composite v0.4 Term 2 substrate-natural decomposition: $N_c^{4/2} N_c = (\text{substrate-color})^{(\text{substrate-Casey-}\#14\text{-codim-4})} / (\text{substrate-Cartan-base})^{(\text{substrate-color})}$; substantive promotion from form-fitting (v0.3) to PARTIAL substrate-mechanism FORCING candidate (v0.4) per Casey #14 STANDING substrate-mechanism cross-link; honest framing per Cal #35 STANDING — substrate-natural factor IDENTIFICATION $\neq$ FORCING DERIVATION; multi-week explicit FORCING derivation per Cal #189.